

Chengzu Li

Webpage <https://chengzu-li.github.io/> [Google Scholar] [Semantic Scholar] GS Citation: 1330 (July 28, 2026)
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Research Interests

Keywords (Multimodal) LLMs, Visual Intelligence, Video Gen., Multimodal Agents, Post-Training.
[*Visual Intelligence & Multimodal Reasoning*] (Visual Planning, ICLR **Oral**; MVoT, ICML);
[*Multimodal Agentic Reasoning*] (UnifiedSKG, EMNLP **Oral**; BINDER, ICLR **Spotlight**);
[*World Models & Video Gen for Planning*] (Thinking in Frames; MSR Cambridge);
[*Evaluation & Capability Understanding*] (TOPVIEWRS, EMNLP **Oral**; 11PLUS-BENCH)

Education

2023 - Now	University of Cambridge • PhD in Computation, Cognition and Language (funded by Cambridge Trust) • Advisor: Prof. Anna Korhonen, Dr. Ivan Vulić (now at DeepMind) and Prof. Serge Belongie.	Cambridge, UK
2022 - 2023	University of Cambridge • M.Phil. in Advanced Computer Science, NLP Track (Distinction 81.7/100) • Advisor: Prof. Simone Teufel. Funded by Cambridge Trust.	Cambridge, UK
2018 - 2022	Xi'an Jiaotong University • Bachelor of Engineering in Automation (91.96/100, top 5%). Minor in Fintech.	Xi'an, China
2021	University of Hong Kong • Exchange student in Computer Engineering (4.08/4.3) • <i>Research Intern</i> , CS Research Internship Programme (Mentors: Tao Yu, Lingpeng Kong)	Hong Kong SAR

Professional Experiences

2026	Google <i>Student Researcher</i> • Mentors: Sercan Ö. Arik • Working on multimodal agents.	London, UK
	Microsoft Research • <i>Research Intern</i> , Game Intelligence, People-Centric AI Group • Mentors: Sergio Valcarcel Macua, Katja Hofmann • Developed world models for planning with video generation in robotics applications.	Cambridge, UK
2024	Microsoft Research <i>Research Intern</i> , GenAI Group • Mentors: Wenshan Wu, Furu Wei • Investigated multimodal reasoning (<i>MVoT</i>) and spatial-reasoning evaluation (<i>11Plus-Bench</i>).	Beijing, China
2023	Toshiba Cambridge Research Laboratory <i>Research Intern</i> • Mentor: Svetlana Stoyanchev, Chao Zhang and Rama Doddipatla • Developed <i>Semantic-Map Based Navigation Instruction Generation</i> .	Cambridge, UK
2021	Shanghai Artificial Intelligence Laboratory <i>Research Intern</i> • Mentor: Lingpeng Kong • Researched structured knowledge grounding in LLMs (<i>UnifiedSKG</i>).	Shanghai, China

Research & Publications

Conferences

(* indicates equal contribution, † indicates corresponding author)

2026

Visual Planning: Let's Think Only with Images

Yi Xu*, Chengzu Li*, Han Zhou*, Xingchen Wan, Caiqi Zhang, Anna Korhonen, Ivan Vulić

ICLR **Oral** [Github]

- Featured at **X(Twitter)**, **Medium** and **etc.(CHN)**

- Community influence with over 350 Github stars

Deforming Videos to Masks: Flow Matching for Referring Video Segmentation

Zanyi Wang, Dengyang Jiang, Liuzhuozheng Li, Sizhe Dang, Chengzu Li, Harry Yang, Guang Dai, Mengmeng Wang, Jingdong Wang

ICLR [Github]

Video Understanding: From Geometry and Semantics to Unified Models

Zhaochong An, Zirui Li, Mingqiao Ye, Feng Qiao, Jiaang Li, Zongwei Wu, Vishal Thengane, Chengzu Li, Lei Li, Luc Van Gool, Guolei Sun, Serge Belongie

Machine Intelligence Research

Thinking in Scales: Accelerating Gigapixel Pathology Image Analysis via Adaptive Continuous Reasoning

Jiusong Ge, Yingkang Zhan, Wenjie Zhao, Di Zhang, Ke Wang, Jiashuai Liu, Chunze Yang, Chengzu Li, Jian Zhang, Yuxin Dong, Ni Zhang, Qidong Liu, Mireia Crispin-Ortuzar, Huazhu Fu, Chen Li, Zeyu Gao

ICML

Reinforcement Learning for Better Verbalized Confidence in Long-Form Generation

Caiqi Zhang*, Xiaochen Zhu*, Chengzu Li, Nigel Collier, Andreas Vlachos

ACL

2025

Information Loss in Vision-Language Models

Wenyan Li, Raphael Tang, Chengzu Li, Caiqi Zhang, Ivan Vulić, Anders Søgaard

EMNLP Findings [Github]

Enriching Patent Claim Generation with European Patent Dataset

Lekang Jiang, Chengzu Li, Stefan Goetz

EMNLP Findings

Imagine While Reasoning in Space: Multimodal Visualization-of-Thought

Chengzu Li*, Wenshan Wu*, Huanyu Zhang, Yan Xia, Shaoguang Mao, Li Dong, Ivan Vulić, Furu Wei

ICML [Github]

- Featured at **IEEE Spectrum**, **TWIML AI Podcast** and **etc.(CHN)**

- Community influence over 70 Github stars

Large Language Models are Miscalibrated In-Context Learners

Chengzu Li, Han Zhou, Goran Glavaš, Anna Korhonen, Ivan Vulić

ACL Findings [Github]

2024

TOPVIEWRS: Vision-Language Models as Top-View Spatial Reasoners

Chengzu Li*, Caiqi Zhang*, Han Zhou, Nigel Collier, Anna Korhonen, Ivan Vulić

EMNLP **Oral** [Github]

- Featured at **TWIML AI Podcast**

Semantic Map-based Generation of Navigation Instructions

Chengzu Li, Chao Zhang, Simone Teufel, Rama Sanand Doddipatla, Svetlana Stoyanchev

COLING

2023

Generating Data for Symbolic Language with Large Language Models

Jiacheng Ye, Chengzu Li, Lingpeng Kong, Tao Yu

EMNLP [Github]

Binding Language Models in Symbolic Languages

Zhoujun Cheng*, Tianbao Xie*, Peng Shi, Chengzu Li, Rahul Nadkarni, Yushi Hu, Caiming Xiong, Dragomir Radev, Mari Ostendorf, Luke Zettlemoyer, Noah A. Smith, Tao Yu

ICLR Spotlight [Github]

2022

UnifiedSKG: Unifying and Multi-Tasking Structured Knowledge Grounding with Text-to-Text Language Models

Tianbao Xie*, Chen Henry Wu*, Peng Shi, Ruiqi Zhong, Torsten Scholak, Michihiro Yasunaga, Chien-Sheng Wu, Ming Zhong, Pengcheng Yin, Sida I. Wang, Victor Zhong, Bailin Wang, Chengzu Li, Connor Boyle, Ansong Ni, Ziyu Yao, Dragomir Radev, Caiming Xiong, Lingpeng Kong, Rui Zhang, Noah A. Smith, Luke Zettlemoyer, Tao Yu

EMNLP Oral [Github]

Pre-prints

From Trainee to Trainer: LLM-Designed Training Environment for RL with Multi-Agent Reasoning

Chao Chen, Chengzu Li[†], Zhiwei Li, Yinhong Liu[†], Zhijiang Guo[†]

Agentic World Modeling: Foundations, Capabilities, Laws, and Beyond

Meng Chu, Xuan Billy Zhang, Kevin Qinghong Lin, ..., Chengzu Li, ..., Ziwei Liu, Philip Torr, Jiaya Jia

How Well Do Models Follow Visual Instructions? VIBE: A Systematic Benchmark for Visual Instruction-Driven Image Editing

Huanyu Zhang*, Xuehai Bai*, Chengzu Li*, Chen Liang, Haochen Tian, Haodong Li, Ruichuan An, Yifan Zhang, Anna Korhonen, Zhang Zhang, Liang Wang, Tieniu Tan

Thinking in Frames: How Visual Context and Test-Time Scaling Empower Video Reasoning

Chengzu Li*, Zanyi Wang*, Jiaang Li*, Yi Xu, Han Zhou, Huanyu Zhang, Ruichuan An, Dengyang Jiang, Zhaochong An, Ivan Vulić, Serge Belongie, Anna Korhonen

Latent Sketchpad: Autoregressive Visual Latent Generation for Interpretable Visual Thoughts in MLLMs

Huanyu Zhang*, Wenshan Wu*, Chengzu Li, Ning Shang, Yan Xia, Yangyu Huang, Yifan Zhang, Li Dong, Zhang Zhang, Liang Wang, Tieniu Tan, Furu Wei

11PLUS-BENCH: Demystifying Multimodal LLM Spatial Reasoning with Cognitive-Inspired Analysis

Chengzu Li, Wenshan Wu, Huanyu Zhang, Qingtao Li, Zeyu Gao, Yan Xia, José Hernández-Orallo, Ivan Vulić, Furu Wei

Scaling and Beyond: Advancing Spatial Reasoning in MLLMs Requires New Recipes

Huanyu Zhang*, Chengzu Li*, Wenshan Wu, Shaoguang Mao, Yifan Zhang, Haochen Tian, Ivan Vulić, Zhang Zhang, Liang Wang, Tieniu Tan, Furu Wei

Academic Service

Paper Reviewing

NLP	ACL ARR 2024, 2025, 2026, COLM 2026
AI	ICLR 2026, ICML 2026, NeurIPS 2026
CV	CVPR 2026, ECCV 2026

Teaching Experiences

2023, 2024	Li18 Computational Linguistics, University of Cambridge Teaching Assistant
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Selected Awards & Grants

2025	Apple AI/ML PhD Fellowship Finalists by University of Cambridge Postgraduate Research Fund from Jesus College, Cambridge Lambda Research Grant (\$2000 USD in GPU credits)
2024	Postgraduate Research Fund from Jesus College, Cambridge Scholar of Jesus College, Cambridge (for outstanding academic performance)
2023	PhD Scholarship from Cambridge Trust (Fully-Funded)
2022	Masters Scholarship from Cambridge Trust (Fully-Funded)
2019	National Scholarship (Top 1%, highest honor for undergraduates in China)

Media Coverage & Talk

2026	Invited Guest Lecture at UCL NLP Course by Ehsan Shareghi on Multimodal Reasoning Invited Talk at Shanghai Artificial Intelligence Laboratory: Visual Planning (ICLR 2026 Oral)
2025	Invited Talk at HKUST (GZ): Reason with Multimodal Minds in Space JiQiZhiXin (CHN): Thinking with Images Alone: RL Fosters a New Paradigm for Reasoning Models XinZhiYuan (CHN): Beyond CoT! Directly Visualize the Multimodal Reasoning Process TWIML AI Podcast with Sam Charrington: Multimodal Visualization-of-Thought IEEE Spectrum: "Thinking" Visually Boosts AI Problem Solving
2024	BMVA: Trustworthy Multimodal Learning with Foundation Models